Digits, Place Value & Base Structure

Integrated learning and diagnostic pack

Digits, Place Value & Base Structure

The central habit

A digit pattern is an **integer written in a representation**. Translate the representation before guessing from examples.

DIGITS -> PLACE VALUE -> ARITHMETIC RESTRICTION -> CHECK

If the final question asks for a number of admissible strings, derive the arithmetic restriction here first; then use counting tools.

1. Digits are coefficients of powers of the base

The decimal numeral **abc** means

$100a+10b+c$.

The base-**b** numeral **$d_k...d_1d_0$** means

$d_k b^k+...+d_1 b+d_0$, with **$0<=d_i<b$** and leading digit nonzero.

This simple translation is the first line for most digit problems.

2. Divisibility by 9 comes from place value

Because **10** leaves remainder 1 on division by 9, every power **10^k** does too. Thus a decimal number and its digit sum have the same remainder modulo 9.

For **abcd**,

$1000a+100b+10c+d$ has the same remainder as **$a+b+c+d$** modulo 9.

The rule is not magic; it is place value collapsing.

3. Divisibility by 11 alternates

Here **10** leaves remainder **-1** modulo 11. Powers alternate: **$10^0,10^1,10^2,...$** behave like **$1,-1,1,-1,...$**.

So **abcd** has the same remainder as

$-a+b-c+d$ modulo 11.

A four-digit palindrome **abba** therefore has alternating sum zero, so it is automatically divisible by 11.

4. Concatenation is block place value

If block **x** is followed by a **k**-digit block **y**, then

$N=10^k x+y$.

Two two-digit blocks: **$100x+y$**. A three-digit block repeated twice: **$1000x+x=1001x$**.

This is usually cheaper than expanding individual digits.

Historical pattern: if a two-digit **p** is followed by a two-digit **q**, then **$N=100p+q$**. If the divisor is **$p+q$**, rewrite

$N=99p+(p+q)$.

Now the divisibility condition is visible.

5. Carries control digit-sum changes

Let **t** be the number of trailing 9s in decimal **n**. Adding 1 turns those **t** nines into zeros and increases the preceding digit by 1. Therefore

$s(n+1)=s(n)+1-9t$.

If **$t=0$**, the digit sum rises by 1. If **$t>=1$**, it may fall sharply.

This replaces blind search with a carry invariant.

6. Digit products are prime-compatibility problems

For a nonzero digit product to be squarefree, no prime may appear twice in the combined prime factorisation of all non-one digits.

Consequences:

- digit 4,8,9 is impossible immediately because it already contains a square prime factor;
- repeating digit 6 is impossible because each 6 contributes factors 2 and 3;
- digits 2 and 6 cannot both appear because prime 2 repeats;

- digit 1 may repeat freely because it changes neither product nor prime exponents.

The squarefree theorem itself belongs to prime-exponent structure; here we use it to restrict digits.

7. Base representation works the same way

352_7 = 3*7^2+5*7+2 = 184.

For an unknown base, the numeral becomes an equation. If **111_b=31**, then

b^2+b+1=31,

so **b=5** after the base condition **b>1** is enforced.

Carries also generalize: in base 8, **777_8+1=1000_8**.

8. Arithmetic restriction vs counting

Suppose place value proves that a four-digit string must satisfy digit sum 12. The arithmetic work is now done. If the question asks **how many** such strings exist, define the admissible strings and count them using the counting toolkit.

Do not turn place-value derivation into a permutations chapter, and do not ask counting to rediscover divisibility rules.

9. Source-anchor recognition

- **abcab** type patterns: write the whole numeral algebraically before testing digits.
- **n** and **n+1** digit sums: count trailing 9s.
- concatenated two-digit blocks: use **100p+q** and reduce against the divisor structure.
- squarefree digit product: classify prime usage before maximizing digit count.

Closing checks

Leading digit nonzero; every digit in the base range; carry length correct; block length correct in concatenation; divisibility condition applied to the value of the numeral; arithmetic restriction separated from any later counting step.

First-Step Reference - Digits, Place Value & Base Structure

Visible clue	First question	First useful line
abc... digit notation	what are place weights?	expand as powers of 10
divisible by 9	what is 10 mod9?	collapse to digit sum
divisible by 11	what is 10 mod11?	alternating signed sum
block appended	how many digits in new block?	**10^k x+y**
n+1 and digit sum	how many trailing 9s?	**s(n+1)=s(n)+1-9t**
squarefree digit product	which prime exponent would repeat?	factor allowed digits conceptually
base-b numeral	what are powers of b?	**sum d_i b^i**
count admissible strings	has arithmetic restriction been derived?	freeze restriction, then count

Close contrasts

- Digit sum vs digit product: additive place-value collapse versus multiplicative prime compatibility.
- Place-value reduction vs generic modular cycle: derive the decimal structure first; retrieve modular legality only as needed.
- Concatenation vs permutation: joining blocks changes numerical value by a power of the base.
- Carry reasoning vs brute force: trailing maximal digits determine the entire digit-sum jump.
- Arithmetic restriction vs string counting: different canonical jobs.

Recognition and First-Line Lab

Write only the first useful mathematical line.

1. Decimal numeral **abcde**.
2. Five-digit palindrome divisible by 9.
3. Six-digit number formed by repeating a three-digit block.
4. A number and its successor have very different digit sums.
5. A nonzero digit product is squarefree.
6. A numeral is given in base 7.
7. A base is unknown but the numeral value is known.
8. A digit restriction has already been derived and only the number of strings is requested.
9. A four-digit numeral must be divisible by 11.
10. A two-digit block **p** followed by **q** is tested for divisibility by **p+q**.

Decide before calculating

A. **123123**: repeated block or digit sum? What is the first representation?

B. **n** ends in 999: carry invariant or generic modular cycle?

C. Product of digits is squarefree: digit-sum rule or prime compatibility?

D. Count five-digit multiples of 9 with restrictions: which part belongs to arithmetic and which to counting?

Practice and Transfer Bank

1. Write the three-digit number abc algebraically in a,b,c.
2. For a four-digit number abcd, express its remainder modulo 9 using its digits.
3. For a four-digit number abcd, express its remainder modulo 11 using digits.
4. Concatenate 37 and 42. What number is formed?
5. Let N be formed by concatenating a three-digit x with a two-digit y. Write N.
6. If s(n)=23 and n ends in exactly two 9s, find s(n+1).
7. If n has no trailing 9 and digit sum 31, find s(n+1).
8. What is the digit sum of 9998+1?
9. How many digits are possible for a nonzero digit d if d itself is squarefree?
10. Find the largest number of occurrences of digit 6 in a numeral whose nonzero digit product is squarefree.
11. A base-7 numeral 352_7 equals what in base 10?
12. Write decimal 45 in base 2.
13. In base b, what is the value of numeral 101_b?
14. A two-digit decimal number 10a+b is divisible by 9. What restriction do the digits satisfy?
15. A four-digit number abba is divisible by 11. Is this automatic?
16. Find the remainder of 123456 modulo 9 using digits.
17. If a two-digit p is followed by the same p, express the four-digit number and factor it.
18. A three-digit block x is repeated twice. Factor the resulting six-digit number.
19. Digit sum changes from 28 to 11 after adding1. How many trailing 9s did n have?
20. Can a numeral with digit product squarefree contain digit 4?
21. Can it contain both digits 2 and 6?
22. In base 5, what decimal number is 243_5?
23. A five-digit decimal palindrome abcba. What is it modulo 9 in terms of digits?
24. Once arithmetic shows digit strings must have digit sum 12, which topic owns counting how many strings satisfy it?

After each problem, name the representation that made the arithmetic restriction visible.

Mastery Check

No method labels or hints are supplied on the first attempt.

1. Find the digit sum of the largest three-digit multiple of 11 whose three digits are pairwise distinct.
2. A number ends in exactly three 9s and has digit sum 40. Find the digit sum after adding 1.
3. A two-digit block p is repeated three times. Express the six-digit number as a multiple of p; give the multiplier.
4. Find decimal value of 314_6.
5. A four-digit number abcd is divisible by both 9 and 11. State the two digit restrictions.
6. Can a number with squarefree nonzero digit product contain both 3 and 6?
7. The digit sum of n is 17 and of n+1 is 9. How many trailing 9s?
8. A three-digit number abc is divisible by 11 and a+c<11. What equality must hold?
9. If decimal N is concatenation of 123 and a two-digit y, write N modulo 123+y.
10. A base-b numeral 111_b has decimal value 31. Find b.
11. Find the smallest positive n whose digit sum decreases when 1 is added.
12. A repeated three-digit block x gives six-digit N. Which fixed integer always divides N?
13. A five-digit palindrome abcba is divisible by 9. Give the restriction modulo9.
14. In base 8, find 777_8+1 and express answer in base8.
15. A numeral has nonzero squarefree digit product and contains digit 5. Which of digits 1,2,3,5,6,7 may also occur without immediately violating squarefreeness?
16. An arithmetic restriction leaves 20 admissible four-digit strings. What should number theory do next if the question asks how many?